

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

MARK SCHEME

CLASS

BIOLOGY

9648/01

Paper 1 Multiple Choice

September 2013

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name and CT on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

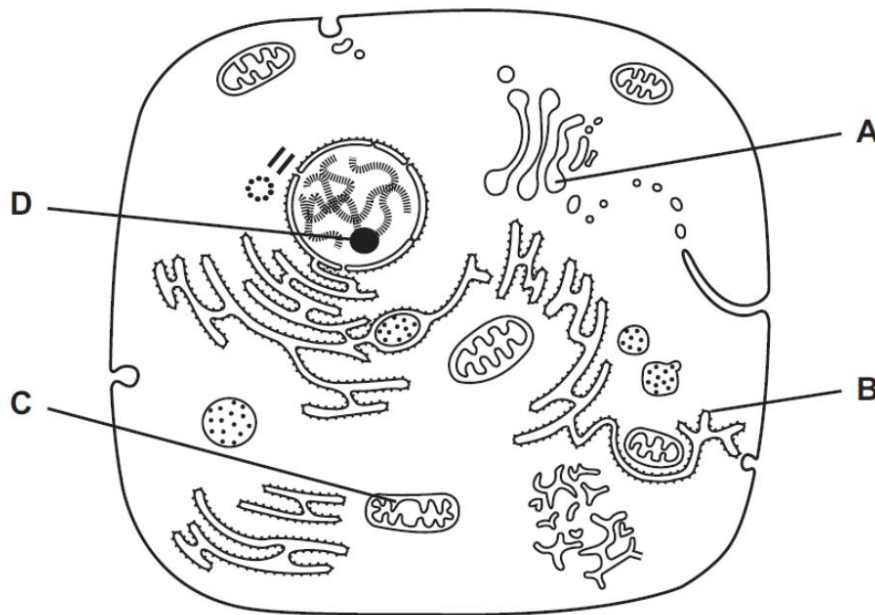
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Calculators may be used.

- 1 The diagram shows the ultrastructure of a eukaryotic cell.

Which organelle does not contain nucleic acid? **A**



- 2 When not involved in protein synthesis, ribosomes exist as separate subunits.

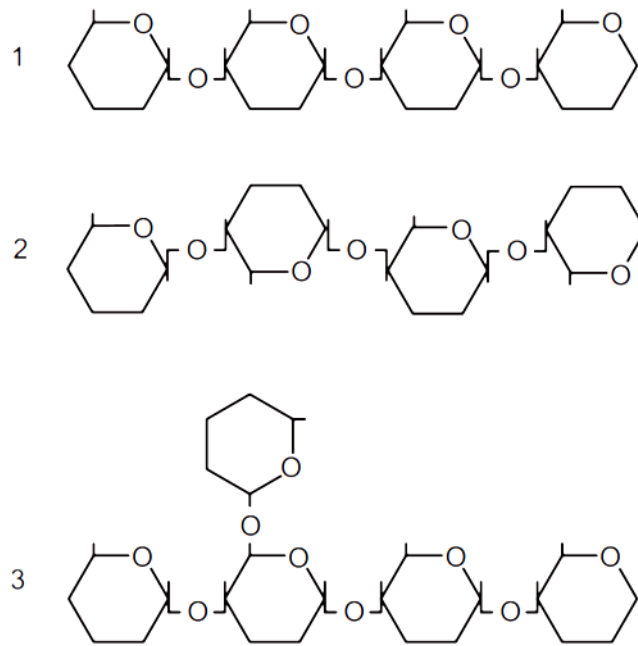
What do these subunits consist of?

- A mRNA and lipid
- B mRNA and tRNA
- C rRNA and lipid
- D rRNA and protein**

- 3 Which statement is true for cellulose, but **not** true for protein?

- A It is found in cell surface membranes.
- B It is synthesised from identical sub-units.**
- C It is used as an energy source.
- D It may be a structural component.

- 4 Diagrams 1, 2 and 3 show the structural formulae of three polysaccharides.

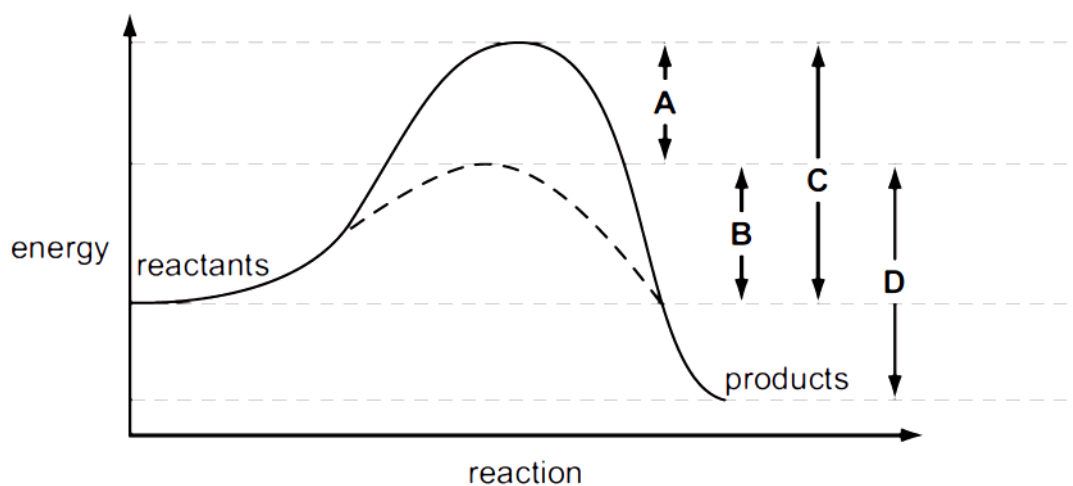


What are the names of these polysaccharides?

	1	2	3
A	amylose	cellulose	Glycogen
B	amylose	glycogen	Amylose
C	cellulose	glycogen	Cellulose
D	glycogen	amylose	Amylose

- 5 The graph shows the activation energy of an enzyme-catalysed reaction and the same reaction without a catalyst.

Which arrow shows the activation energy of the uncatalysed reaction? **C**



6 Which of the following statement about chromosome structure is true?

- 1 Linker DNA refers to the linear double-stranded DNA between adjacent nucleosomes.
- 2 The 30 nm chromatin fibre has a solenoid structure.
- 3 The 300 nm chromatin fibre is attached to multiple locations on a central protein scaffold.
- 4 Euchromatin is the more diffuse region of the interphase genetic material and is transcriptionally inactive.

A 1 and 2

B 1 and 4

C 2 and 3

D 3 and 4

7 If there were 30 amino acids and DNA only contained three types of nitrogenous bases, what would be the minimum number of bases per codon that could code for proteins?

A 3

B 4

C 6

D 8

8 What occurs as a result of base pairing during DNA replication?

A Both strands of the DNA have the same sequence of purines and pyrimidines.

B In each strand, the number of purine bases equals the number of pyrimidine bases.

C Purine bases in one strand become hydrogen-bonded to purine bases in the other strand.

D The number of purine bases in one strand equals the number of pyrimidine bases in the other strand.

- 9 Part of the amino acid sequences in normal and sickle cell haemoglobin are shown.

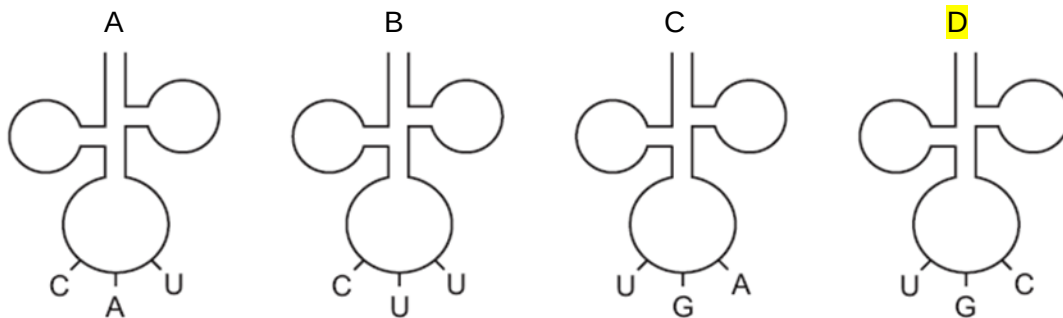
normal haemoglobin
thr-pro-glu-glu

sickle cell haemoglobin
thr-pro-val-glu

Possible mRNA codons for these amino acids are:

glutamate (glu) GAA GAG proline (pro) CCU CCC
threonine (thr) ACU ACC valine (val) GUA GUG

Which tRNA molecule is **not** involved in the formation of this part of the sickle cell haemoglobin?



- 10 Which of the following is characteristic of the lytic cycle?

- A Many bacterial cells containing viral DNA are produced.
- B Viral DNA is incorporated into the host genome.
- C The viral genome replicates without destroying the host.
- D A large number of phages are released at a time.**

- 11 A bacterium is infected with an experimentally constructed bacteriophage composed of the T2 phage protein coat and T4 phage DNA. The new phages produced would have _____.

- A T2 protein and T4 DNA
- B T2 protein and T2 DNA
- C T4 protein and T4 DNA**
- D a mixture of the DNA and proteins of both phages

12 Which statement about the genomes of prokaryotes is correct?

- A Prokaryotic chromosomes are sometimes called plasmids.
- B Prokaryotic cells have multiple chromosomes, "packed" with a relatively large amount of protein.
- C Prokaryotic genomes are composed of linear DNA (that is, DNA existing in the form of a line with two ends).
- D The prokaryotic chromosome is not contained within a nucleus but, rather, is found at the nucleoid region.**

13 If the experimental population of *E. coli* lacks an F factor or F plasmid, and if bacteriophage are excluded from the bacterial cultures, then which of these is a means by which beneficial mutations might be transmitted horizontally to other *E. coli* cells?

- A via conjugation
- B via transduction
- C via transformation**
- D both A and C above

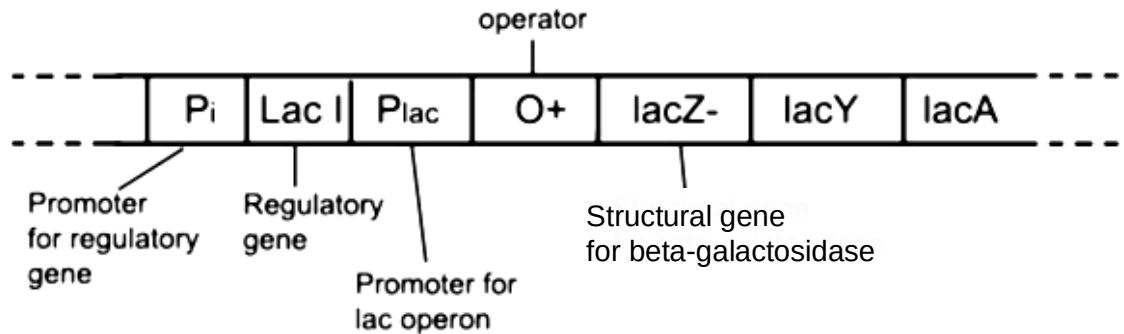
14 The lactose operon is likely to be transcribed when _____.

- A there is more glucose in the cell than lactose
- B there is glucose but no lactose in the cell
- C the cyclic AMP and lactose levels are both high within the cell**
- D the cAMP level is high and the lactose level is low

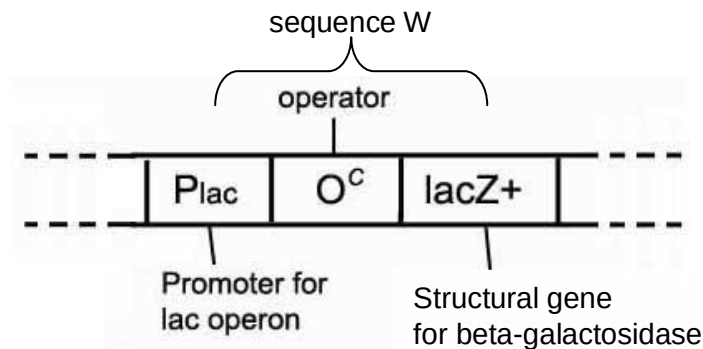
15 How is the *trp* operon different from the *lac* operon?

	<i>trp</i> operon	<i>lac</i> operon
A	Produces polycistronic mRNA	Produces monocistronic mRNA
B	Repressor binds to operator	Activator binds to operator
C	Repressible	Inducible
D	Transcription occurs in the presence of tryptophan	Transcription occurs in the absence of glucose

- 16 The diagram below depicts a lac operon of the wild type bacterial chromosome in *E. coli*. The *lacZ* gene is regulated via the normal operator sequence (O^+). However, the *lacZ* allele is a recessive form (*lacZ*⁻) that does not code for beta-galactosidase.



This particular *E. coli* strain is conjugated with a mutant strain of *E. coli* carrying the P_{lac} O^c *lacZ*⁺ DNA sequence as shown below. In this sequence, the *lacZ* allele is normal (*lacZ*⁺) but the operator (O) is mutated. The O^c mutant is unable to bind the lac repressor coded for by *lacI*.



How would the expression of the *lacZ* gene be regulated in the recipient cells after homologous recombination between the wild type (recipient) and the mutant strain (donor) involving sequence **W**?

- A There will be normal regulation of lactose metabolism.
 - B There will be continuous expression of beta-galactosidase regardless of glucose level.**
 - C The inability to synthesise the *lacZ* gene product, beta-galactosidase.
 - D The *lacZ* gene is inducible, but unable to be repressed by high glucose levels.
- 17 Which of the following statements concerning proto-oncogenes is **false**?
- A They can code for proteins associated with cell growth.
 - B They are produced by somatic mutations induced by carcinogenic substances.**
 - C They can be involved in producing proteins for cell adhesion.
 - D They can code for proteins involved in cell division.

18 Which of the following event(s) is/are necessary for the production of a malignant tumour?

- 1 activation of an oncogene in the cell
- 2 inactivation of tumour-suppressor genes within the cell
- 3 presence of mutagenic substances within the cell's environment
- 4 presence of a retrovirus within the cell

A 1 and 2

B 1 only

C 3 only

D 3 and 4

19 In *Drosophila* long wing and broad abdomen are dominant to vestigial wing and narrow abdomen. A pure-breeding long-winged, broad-abdomened *Drosophila* is crossed with pure-breeding vestigial-winged and narrow-abdomened *Drosophila*. Two of the F1 generation were crossed and the following results were produced in the F2 generation.

long wing, broad abdomen	482
vestigial wing, narrow abdomen	154

Which one of the following is illustrated by these results?

A autosomal linkage

B incomplete dominance

C Mendelian dihybrid inheritance

D Mendelian monohybrid inheritance

20 In cats, the genes controlling coat-colour are co-dominant and carried on the X chromosomes. When a black female was mated with a ginger male the resulting litter consisted of black male and tortoise-shell female kittens.

What phenotypic ratio would be expected in the F2 generation?

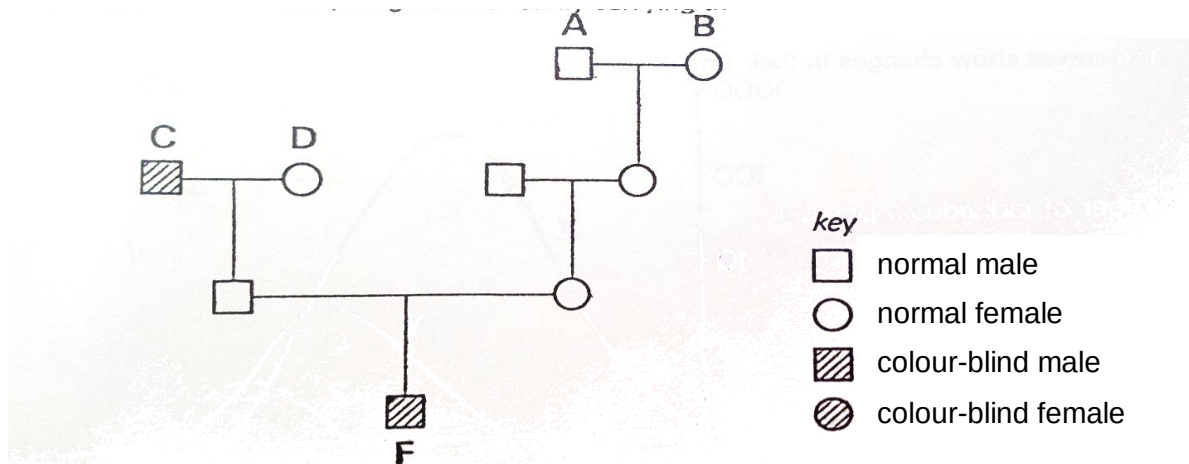
A 1 black male : 1 ginger male : 2 tortoise-shell females

B 1 black male : 1 ginger male : 1 tortoise-shell female : 1 black female

C 2 black males : 1 tortoise-shell female : 1 ginger female

D 2 black males : 1 tortoise-shell female : 1 black female

- 21 The diagram shows the pedigree of a family carrying sex-linked allele for colour-blindness.



From which labelled member of his family did F inherit this disorder? **B**

- 22 In guinea pigs, the allele R for rough coat is dominant over allele r for smooth coat and the allele B for black fur is dominant over the allele b for white fur. The genes for fur colour and texture are not linked.

Two guinea pigs with genotype RrBb were mated together and one of the offspring had a rough and black coat.

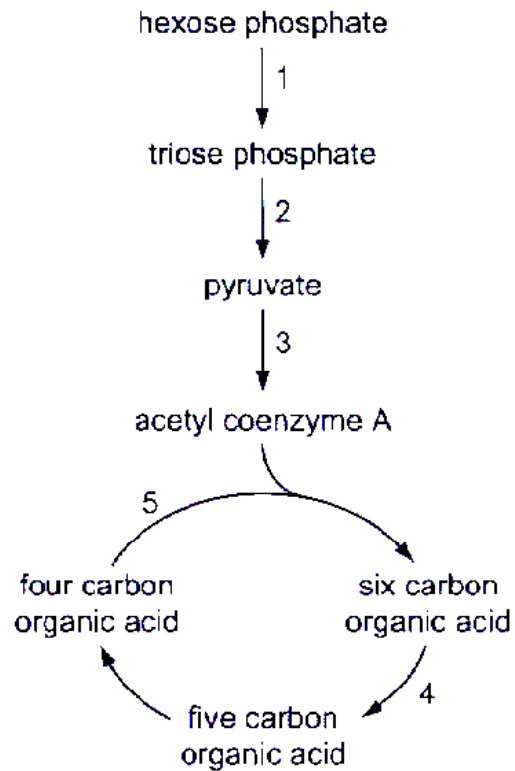
What is the probability that this offspring was homozygous for both rough coat and black fur?

- A 1 in 16
B 1 in 9
 C 2 in 16
 D 2 in 9

- 23 Which combination would decrease the fluidity of a cell surface membrane?

	Saturated fatty acids	Non-saturated fatty acids	Cholesterol
A	decrease	increase	decrease
B	decrease	increase	increase
C	increase	decrease	decrease
D	increase	decrease	increase

24 The flow diagram shows some of the stages in respiration.



During which stages does oxidative decarboxylation occur?

- A 1 and 3
- B 2 and 3
- C 3 and 4**
- D 4 and 5

25 Six tubes containing preparations from animal tissue were set up as shown in the table.

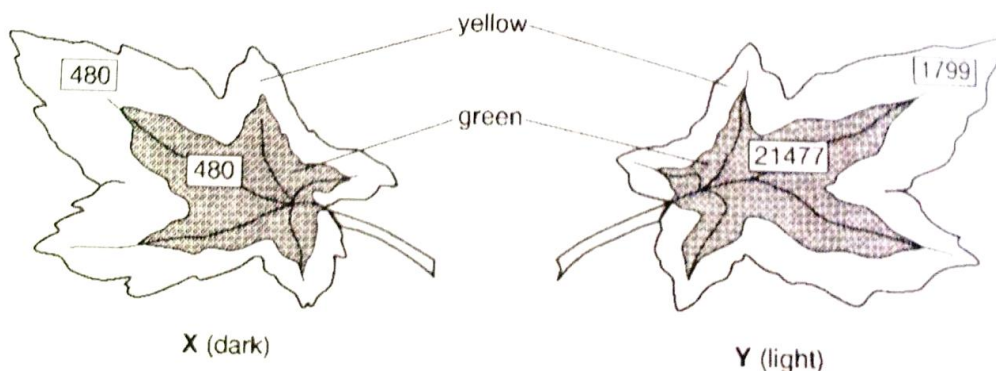
tube	Contents
1	glucose + homogenised cells
2	glucose + mitochondria
3	glucose + cytoplasm lacking organelles
4	pyruvate + homogenised cells
5	pyruvate + mitochondria
6	pyruvate + cytoplasm lacking organelles

After incubation, in which three tubes would carbon dioxide be produced?

- A 1, 2 and 4
- B 1, 4 and 5**
- C 2, 4 and 5
- D 2, 4 and 6

26 Variegated leaves of a plant were supplied with radioactive carbon dioxide ($^{14}\text{CO}_2$) during an experiment. Leaf X was kept in the dark and leaf Y was kept in the light.

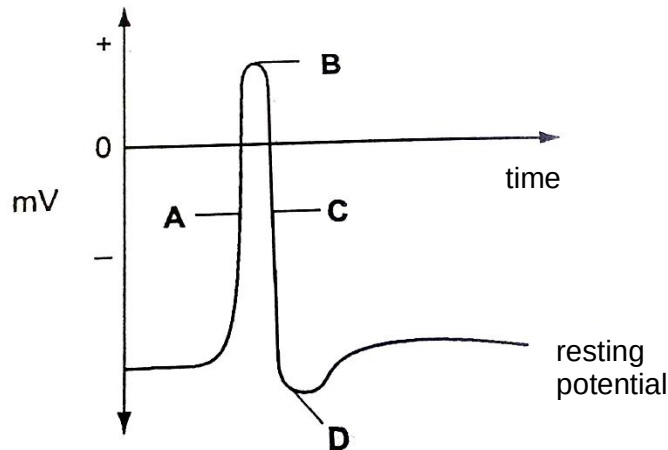
At the end of the experiment, the radioactivity in the leaves was measured. The results (in arbitrary units) are shown in the boxes in the diagram



What is the most likely explanation for the level of radioactivity found in the yellow zone of leaf Y?

- A Photosynthesis occurs but no storage of starch occurs in this zone.
- B Photosynthesis proceeds slowly in the absence of chlorophylls a and b.
- C Products of photosynthesis are transported into the yellow zone.**
- D Radioactive carbon dioxide diffuses into the leaf and accumulates there.

27 The diagram shows an action potential.



At which point on the graph is the membrane most permeable to sodium ions? **A**

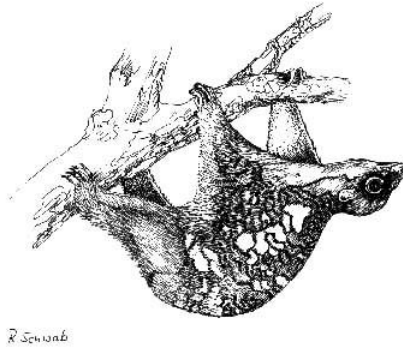
28 What effect do sexual processes (meiosis and fertilization) have on the allelic frequencies in a population?

- A** They tend to reduce the frequencies of deleterious alleles and increase the frequencies of advantageous ones.
- B** They tend to increase the frequencies of deleterious alleles and reduce the frequencies of advantageous ones.
- C** They tend to selectively combine favourable alleles into the same zygote but do not change allelic frequencies.
- D** They have no effect on allelic frequencies.

29 As adults, certain species of whales possess baleen instead of teeth. Baleen is used to filter the whales' diet of planktonic animals from seawater. As embryos, baleen whales possess teeth, which are later replaced by baleen. The teeth of embryonic baleen whales are evidence that

- A** all whales are descendents of terrestrial mammals.
- B** baleen whales are descendents of toothed whales.
- C** baleen embryos pass through a stage when they resemble adult toothed whales.
- D** among ancient whales, baleen evolved before teeth.

- 30 The population of the Malayan Colugo in Singapore shown below are descendants from a small population of Colugos that were over-hunted and had their habitats destroyed post-1819 when 90% of primary forests in Singapore were cleared.



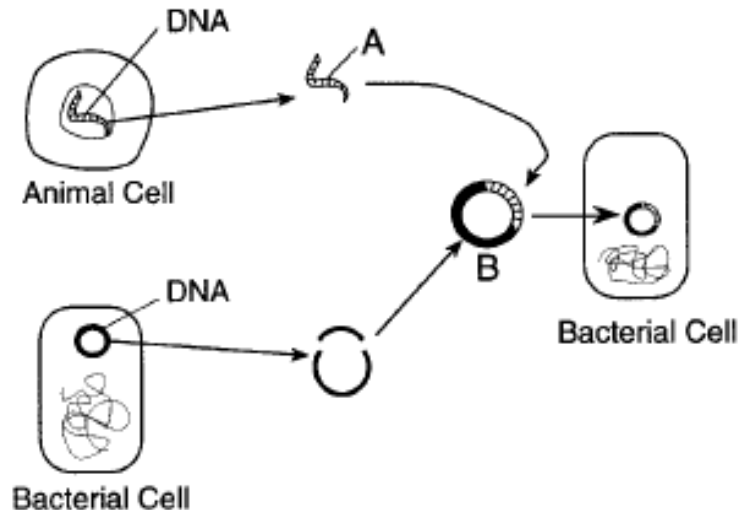
Genetic variability in this population would be expected to be _____ due to _____.

- A slight, a population bottleneck
 - B slight, the founder effect
 - C great, disruptive selection
 - D great, a population bottleneck
- 31 In using the average mutation rate of a gene as a molecular clock, which of the following would act to make the clock underestimate the true amount of time that had elapsed since two related species diverged from their common ancestor?

A mutation

- A to the second base of a nucleotide codon.
 - B that is synonymous (a.k.a. a silent mutation).
 - C that returns a mutated nucleotide to its original state.
 - D substitutes a different amino acid in place of the original amino acid.
- 32 Which of the following statements about restriction enzymes is **false**?
- A They are protein in nature.
 - B They are naturally produced by some organisms.
 - C Different restriction enzymes recognize and cut at different sites.
 - D Restriction enzymes are not involved in establishing a cDNA library.

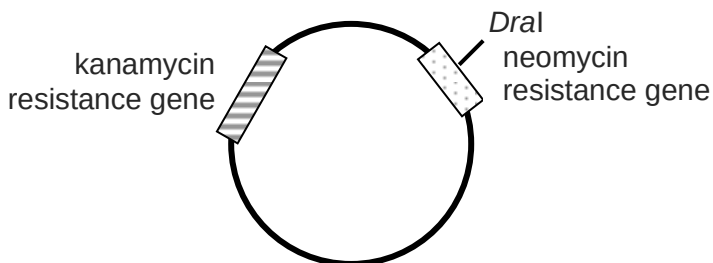
- 33 The diagram below shows the general procedure in the cloning of an animal gene to study gene expression.



Which of the following is true of structures A and B?

	Structure A contains	Structure B contains
A	a genetic code	an origin of replication
B	a DNA probe	selectable markers
C	an origin of replication	Ribonucleotides
D	Ribonucleotides	enhancer sequences

- 34 The diagram below shows a plasmid carrying two selectable markers which code for resistance to kanamycin and neomycin. The restriction site is also shown.



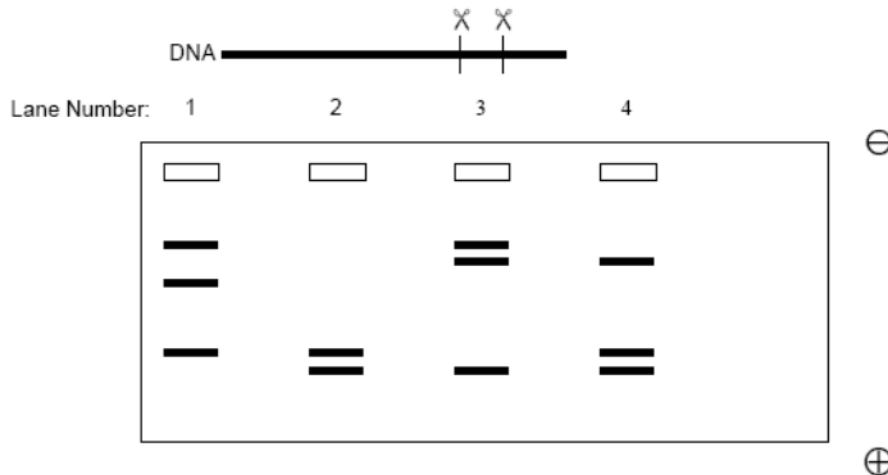
A eukaryotic gene is produced by the restriction endonuclease *Dral*. This gene was inserted and ligated into a cut plasmid containing compatible ends. The *E. coli* transformation mixture was grown in four different media: nutrient medium plus kanamycin, nutrient medium plus neomycin, nutrient medium plus kanamycin and neomycin, and nutrient medium containing no antibiotics.

The bacteria which has taken up the religated plasmid would grow in

- A** nutrient medium containing kanamycin only.
- B** nutrient medium containing neomycin only.
- C** nutrient medium containing both kanamycin and neomycin.
- D** all four types of medium.

35 A sample of DNA is cut at the positions indicated (marked with X) below.

When the fragments are loaded and run on an electrophoresis gel, which lane (1 to 5) has the pattern that you would expect?



- A 1
- B 2
- C 3
- D 4**

36 Southern blot can be used to detect all of the following **except**

- A whether or not a gene is being expressed.**
- B the presence/absence of a specific DNA sequence.
- C the size of restriction fragment that encodes a particular DNA sequence.
- D the differences in DNA sequences from different organisms.

37 The following are advantages of transgenics **except**:

- A A gene for a desirable characteristic can be identified and cloned.
- B All the beneficial characteristics of an existing variety can be kept and just the desired new gene can be added.
- C Sexual reproduction is necessary.**
- D Transgenics is much faster than conventional breeding.

38 What are the normal functions of stem cells in a living human?

	Functions		
A	differentiating into many kinds of cells in a 3 -5 day old human embryo	producing insulin in pancreas damaged by type 1 diabetes	cells that can differentiate into bone cells in the skeleton
B	differentiating into many kinds of cells in a 3 -5 day old human embryo	producing red blood cells worn out by normal wear and tear	cells that can differentiate into cardiac muscles in the heart
C	differentiating into only one kind of cells in a 3 -5 day old human embryo	producing dopamine in brains of people with Parkinson's disease	cells that can differentiate into cartilage cells in the joint
D	differentiating into only one kind of cells in a 3 -5 day old human embryo	producing differentiated cells that can be used to screen new drugs	cells that can differentiate into nerve cells in the brain

39 Which of the following disorders would gene therapy be **least** effective?

- A Cystic fibrosis
- B Haemophilia
- C Huntington's disease**
- D Sickle cell anaemia

40 What are some similarities between all cancer cells and all stem cells?

- A They replicate indefinitely, are non-differentiated and can be found in various parts of the body.**
- B They replicate indefinitely, lack cell-cell adhesion and are able to move from one location in the body to another.
- C They lack contact inhibition, are non-differentiated and are regulated by molecular signals.
- D They are able to move from one location in the body to another, are regulated by molecular signals and can be found in various parts of the body.